

Algebraic Expressions

Adding and Subtracting

$$\textcircled{1} 2x + 7x = \boxed{9x}$$

$$\textcircled{2} 11x - 5x = \boxed{6x}$$

$$\textcircled{3} 4x - 6x = \boxed{-2x}$$

$$\textcircled{4} -5x + x = \boxed{-4x}$$

$$\textcircled{5} -3x + 8x = \boxed{5x}$$

$$\textcircled{6} 3y + (-6y) = \boxed{-3y}$$

$$\textcircled{7} 3y - 6y = \boxed{-3y}$$

$$\textcircled{8} -10x + 5x = \boxed{-5x}$$

$$\textcircled{9} (-10x) + 5x = \boxed{-5x}$$

$$\textcircled{10} (-2x) + (-9x) = \boxed{-11x}$$

$$\textcircled{11} -2x - 9x = \boxed{-11x}$$

$$\textcircled{12} -4x - 3x = \boxed{-7x}$$

Double Negatives

$$\textcircled{13} 7x - (-5x) = 7x + 5x = \boxed{12x}$$

$$\textcircled{14} -2x - (-4x) = -2x + 4x = \boxed{2x}$$

$$\textcircled{15} 4x + 5x = \boxed{9x}$$

$$\textcircled{16} 3x - 8x = \boxed{-5x}$$

$$\textcircled{17} 2x - 10x = \boxed{-8x}$$

$$\textcircled{18} -4x + 2x = \boxed{-2x}$$

$$\textcircled{19} -x + 7x = \boxed{6x}$$

$$\textcircled{20} 6y + (-6y) = \boxed{0}$$

$$\textcircled{21} 8y - 11y = \boxed{-3y}$$

$$\textcircled{22} -4x + 3x = \boxed{-x}$$

$$\textcircled{23} (-4x) + 3x = \boxed{-x}$$

$$\textcircled{24} (-5x) + (-3x) = \boxed{-8x}$$

$$\textcircled{25} -5x - 3x = \boxed{-8x}$$

$$\textcircled{26} -3x - 9x = \boxed{-12x}$$

Double Negatives

$$\textcircled{27} 3y - (-y) = 3y + y = \boxed{4y}$$

$$\textcircled{28} -6x - (-3x) = -6x + 3x = \boxed{-3x}$$

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$$\textcircled{29} 4x + 9x = \boxed{13x}$$

$$\textcircled{30} 2x - 5x = \boxed{-3x}$$

$$\textcircled{31} x - 16x = \boxed{-15x}$$

$$\textcircled{32} -9x + 2x = \boxed{-7x}$$

$$\textcircled{33} -4y + 10y = \boxed{6y}$$

$$\textcircled{34} 7x + (-x) = \boxed{6x}$$

$$\textcircled{35} 11x - 12x = \boxed{-x}$$

$$\textcircled{36} -5x + 11x = \boxed{6x}$$

$$\textcircled{37} (-5x) + 11x = \boxed{6x}$$

$$\textcircled{38} (-7x) + (-8x) = \boxed{-15x}$$

$$\textcircled{39} -7x - 8x = \boxed{-15x}$$

$$\textcircled{40} -3x - 5x = \boxed{-8x}$$

Double Negatives

$$\textcircled{41} 10y - (-3y) = 10y + 3y = \boxed{13y}$$

$$\textcircled{42} -8y - (-11y) = -8y + 11y = \boxed{3y}$$

$$\textcircled{43} 6x - 14x + 3x \\ -8x + 3x \\ \boxed{-5x}$$

$$\textcircled{44} x + (-6x) + 2x \\ -5x + 2x \\ \boxed{-3x}$$

$$\textcircled{45} -8z + 9z - (-2z) \\ z - (-2z) \\ z + 2z \\ \boxed{3z}$$

$$\textcircled{46} -5x - 10x + 4x \\ -15x + 4x \\ \boxed{-11x}$$

$$\textcircled{47} (-3x) + (-12x) + (-2x) \\ -15x + (-2x) \\ \boxed{-17x}$$

$$\textcircled{48} -3x - 12x - 2x \\ -15x - 2x \\ \boxed{-17x}$$

$$\textcircled{49} -2x + 4x + (-8x) \\ 2x + (-8x) \\ \boxed{-6x}$$

$$\textcircled{50} 6z - (-3z) - 9z \\ 6z + 3z - 9z \\ 9z - 9z \\ \boxed{0}$$

$$\textcircled{51} -4x + (-2x) + x \\ -6x + x \\ \boxed{-5x}$$

$$\begin{array}{r} (52) -3x - 6x + 4x \\ -9x + 4x \\ \hline -5x \end{array}$$

$$\begin{array}{r} (53) 2x - 8x - 9x \\ -6x - 9x \\ \hline -15x \end{array}$$

$$\begin{array}{r} (54) (-4x) + (-3x) + (-8x) \\ -7x + (-8x) \\ \hline -15x \end{array}$$

$$\begin{array}{r} (55) -4x - 3x - 8x \\ -7x - 8x \\ \hline -15x \end{array}$$

$$\begin{array}{r} (56) 2x + (-x) + 9x \\ x + 9x \\ \hline 10x \end{array}$$

$$\begin{array}{r} (57) 8z - 10z + 2z \\ -2z + 2z \\ \hline 0 \end{array}$$

$$\begin{array}{r} (58) -3z + (-4z) - (-11z) \\ -3z + (-4z) + 11z \\ -7z + 11z \\ \hline 4z \end{array}$$

$$\begin{array}{r} (59) (-9z) + 2z + (-10z) \\ -7z + (-10z) \\ \hline -17z \end{array}$$

$$\begin{array}{r} (60) (-2x) + (-3x) + (-x) \\ -5x + (-x) \\ \hline -6x \end{array}$$

$$\begin{array}{r} (61) -2x - 3x - x \\ -5x - x \\ \hline -6x \end{array}$$

$$\begin{array}{r} (62) 7x - 4x + (-2x) + (-3x) \\ 3x + (-2x) + (-3x) \\ x + (-3x) \\ \hline -2x \end{array}$$

$$\begin{array}{r} (63) -11x - 4x + 8x - (-x) \\ -15x + 8x - (-x) \\ -7x - (-x) \\ -7x + x \\ \hline -6x \end{array}$$

$$\begin{array}{r} (64) -7x + (-5x) - 8x + 2x \\ -12x - 8x + 2x \\ -20x + 2x \\ \hline -18x \end{array}$$

$$\begin{array}{r} (65) -3x - 6x + 4x - 10x \\ -9x + 4x - 10x \\ -5x - 10x \\ \hline -15x \end{array}$$

$$\begin{array}{r} (66) (-8x) + (-x) + (-2x) + (-4x) \\ -9x + (-2x) + (-4x) \\ -11x + (-4x) \\ \hline -15x \end{array}$$

$$\begin{array}{r} (67) -8x - x - 2x - 4x \\ -9x - 2x - 4x \\ -11x - 4x \\ \hline -15x \end{array}$$

$$x \cdot x = x^2 \quad x^2 \neq 2x$$

Multiplying

$$(68) (-3x^2)6x^7 = \boxed{-18x^9}$$

$$(69) 6y^3(-7y) = \boxed{-42y^4}$$

$$(70) -8x^2y^5(2x^3y^6) = \boxed{-16x^5y^{11}}$$

$$(71) -3x^2y^4z^5(-4x^3y^9z^6) = \boxed{12x^5y^{13}z^{11}}$$

$$(72) (-6y^2z^6x)(-9x^2y^8z^2) = \boxed{54y^{10}x^3z^8}$$

$$(73) -2x^4(5x^3)(-10x^7) = -10x^7(-10x^7) = \boxed{100x^{14}}$$

$$(74) 3x^6y^2(-4y^7)(-6y^9x^{11}) = \boxed{72x^{17}y^{17}}$$

$$(75) 5y^4(-3x^2y)(2x^5y^8) = \boxed{-30y^{13}x^7}$$

$$(76) -9z^3y^5x^2(-x^2y^4z^6)3x^4y^7z^3 = \boxed{27z^{12}y^{16}x^8}$$

$$(77) 4x^2(-y)2x(-3z^3x)(2yz) = \boxed{48x^4y^2z^4}$$

$$(78) (-5x^4)3x^5 = \boxed{-15x^9}$$

$$(79) 6x(-8x^5) = \boxed{-48x^6}$$

$$(80) -2x^5y^7(3x^4y^2) = \boxed{-6x^9y^9}$$

$$(81) -5x^3y^8(-4x^4y) = \boxed{20x^7y^9}$$

$$(82) (-3x^5yz^3)(-8xy^2z^6) = \boxed{24x^6y^3z^9}$$

$$(83) -3xz^3(2x^2y^4)(-x^4yz^7) = \boxed{6x^7y^5z^{10}}$$

$$(84) 4x^3y(-3y)(-5x^9y^6) = \boxed{60x^{12}y^8}$$

$$85) 2x^4y^3(-2)3y^5x^7 = \boxed{-12x^{11}y^8}$$

$$86) -2z^3(-x^7y^2z^4)2x^5z^7 = \boxed{4z^{14}x^{12}y^2}$$

$$87) 3x^2y(-1)2x^5y^7(-3z^8)(2y^5z^9) = \boxed{36x^7y^{13}z^{17}}$$

Dividing

$$88) \frac{-20x^4}{5x^2} = \boxed{-4x^2}$$

$$89) \frac{18x^3}{-28x^7} = \boxed{-\frac{9}{14x^4}}$$

$$90) \frac{-22x^7y^5}{-8x^5y^8} = \boxed{\frac{11x^2}{4y^3}}$$

$$91) \frac{-6x^3y^2}{12xy^2} = \boxed{-\frac{x^2}{2}}$$

$$92) \frac{40x^4yz^9}{-8x^3y^2z^4} = \boxed{\frac{-5xz^5}{y}}$$

$$93) \frac{-28x^5y^8z^5}{-63x^2y^{10}z^6} = \boxed{\frac{4x^3}{9y^2z}}$$

$$94) \frac{-40x^7}{5x^5} = \boxed{-8x^2}$$

$$95) \frac{-28x^4}{-30x^7} = \boxed{\frac{14}{15x^3}}$$

$$96) \frac{-28x^9y^4}{-4xy^6} = \boxed{\frac{7x^8}{y^2}}$$

$$\frac{\begin{array}{c} x \cdot x \cdot x \\ x \cdot x \cdot x \cdot x \cdot x \cdot x \end{array}}{x \cdot x \cdot x \cdot x \cdot x \cdot x}$$

$$97) \frac{-12x^7y^{10}}{15x^8y^2} = \boxed{\frac{4y^8}{5x}}$$

$$98) \frac{77x^7y^8z^4}{-11x^3y^{11}z^2} = \boxed{-\frac{7x^4z^2}{y^3}}$$

$$99) \frac{-25xy^8z^5}{-35x^3yz^6} = \boxed{\frac{5y^7}{7x^2z}}$$

$$100) \frac{-6x^3y^4}{48x^5y^8} = \boxed{-\frac{1}{8x^2y^4}}$$

$$101) \frac{2x^5y^{10}}{-8x^6y^{12}} = \boxed{-\frac{1}{4xy^2}}$$

$$102) \frac{-3x^4y^3}{-9x^7y^5} = \boxed{\frac{1}{3x^3y^2}}$$

$$103) \frac{-10x^8y^{10}}{50x^8y^{13}} = \boxed{-\frac{1}{5y^3}}$$

$$104) \frac{-24x^3y^5}{54x^4y^5} = \boxed{-\frac{4}{9x}}$$

$$105) \frac{-14x^{10}y}{-7x^{10}y^2} = \boxed{\frac{2}{y}}$$

$$106) \frac{6x^3y^9}{-42x^5y^9} = \boxed{-\frac{1}{7x^2}}$$

$$107) \frac{-72x^3y^4}{-81x^3y^4} = \boxed{\frac{8}{9}}$$

$$108) \frac{-45x^8y^7z^3}{5x^8y^7z^3} = \boxed{-9}$$

$$109) \frac{-30x^3y^8z}{-6x^3y^8z} = \boxed{5}$$

$$110) \frac{14x^7yz}{-21x^7yz} = \boxed{-\frac{2}{3}}$$

$$111) 36x^5y^7 \div (-9x^3y) = \boxed{-4x^2y^6}$$

$$112) -35xy \div (-56xy^3) = \boxed{\frac{5}{8y^2}}$$

$$2 \div 3 = \frac{2}{3}$$

